CONSTRUCTIBLE SUBSETS OF A NOETHERIAN SPACE ARE OPEN IN THE CONSTRUCTIBLE TOPOLOGY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterization of constructible subsets as finite unions of locally closed subsets. Invoked inside the counted proof (source lines 3177-3185). Writes the subset as a finite union of pieces whose openness in the constructible topology is easier to see.

Source reference. Original lemma lemma-characterize-constructible-Noetherian: sources/stacks-topology/pinned-original-source.tex, source lines 3113–3113. Navigation pointer only: the counted text refers to the Stacks item lemma-characterize-constructible-Noetherian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a Noetherian topological space. Let $x \in X$. Let $E \subset X$ be constructible in X. The following are equivalent:*

1. *E is a neighbourhood of x, and*
2. *for every irreducible closed subset Y of X which contains x the intersection $E \cap Y$ is dense in Y.*

Proof. It is clear that (1) implies (2). Assume (2). Consider the set $\mathcal{S}$ of closed subsets Y of X containing x such that $E \cap Y$ is not a neighbourhood of x in Y. If $\mathcal{S} \neq \emptyset$, then it has a minimal element Y as X is Noetherian. Suppose $Y = Y_1 \cup Y_2$ with two smaller nonempty closed subsets Y_1, Y_2. If $x \in Y_i$ for $i = 1, 2$, then $Y_i \cap E$ is a neighbourhood of x in Y_i and we conclude $Y \cap E$ is a neighbourhood of x in Y which is a contradiction. If $x \in Y_1$ but $x \notin Y_2$ (say), then $Y_1 \cap E$ is a neighbourhood of x in Y_1 and hence also in Y, which is a contradiction as well. We conclude that Y is irreducible closed. By assumption (2) we see that $E \cap Y$ is dense in Y. Thus $E \cap Y$

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contains an open V of Y, see Lemma lemma-characterize-constructible-Noetherian. If $x \in V$ then $E \cap Y$ is a neighbourhood of x in Y which is a contradiction. If $x \notin V$, then $Y' = Y \setminus V$ is a proper closed subset of Y containing x. By minimality of Y we see that $E \cap Y'$ contains an open neighbourhood $V' \subset Y'$ of x in Y'. But then $V' \cup V$ is an open neighbourhood of x in Y contained in E, a contradiction. This contradiction finishes the proof of the lemma. $\square$